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## PAPER\_559: Buoyancy-Stratified Factorial Geometry — Einstein Tensor and Self-Sourced Field Equations

**Author:** Daniel T. Murphy — Star Magic / UQFF Framework

**Session:** 149 | **Source:** Composed from CP4 #149, #43, #66 (Sessions 148, 107–110)

**CP4 Class:** BSFGEinsteinTensorFieldEquationsCalculator (#154)

**Date:** 2026-03-27

**Context note:** PAPER\_554 (CP4 #149) derived the Riemann curvature  $R^r_{0r0}$  and Ricci scalar  $R_{\text{scalar}}$  for the BSFG metric  $A_{\mu\nu}$ . This paper takes the next step: forming the Einstein tensor  $G_{\mu\nu}$  and establishing the BSFG self-sourced field equations. The central finding — that the BSFG amplification factor  $\text{amp} \gg 1$  — shows that the Aether-metric perturbation does not obey the standard Einstein equations, but instead a stronger BSFG self-sourcing relation.

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### Abstract

This paper presents a UQFF analysis of Stratified Factorial Geometry — Einstein Tensor and Self-Sourced Field Equations, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

### §1 Abstract

We derive the Einstein tensor  $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}A_{\mu\nu}R_{\text{scalar}}$  for the Buoyancy-Stratified Factorial Geometry metric  $A_{\mu\nu}(r) = g_{\mu\nu} + \varepsilon(r)\delta_{\mu\nu}$ , and compare it to the natural Aether source  $\kappa_E T_{s00}$  via standard Einstein equations. The key result is:

$$\text{amp} \equiv \frac{G_{00}}{\kappa_E T_{s00}} \approx \frac{18\eta c^4}{8\pi G r^2}$$

At the solar surface:  $\text{amp} \approx 1.8 \times 10^4$ . This amplification means the BSFG metric perturbation  $\varepsilon = \eta T_{s00} \cos(\pi t_n)$  generates curvature geometrically out of proportion to any local stress-energy source — a hallmark of a non-Einstein field theory. The effective cosmological constant is  $\Lambda_{\text{eff}} = \kappa_E \eta T_{s00}/2 \approx 1.3 \times 10^{-45} \text{ m}^{-2}$ , some seven orders of magnitude above the observed  $\Lambda_{\text{obs}} = 1.1 \times 10^{-52} \text{ m}^{-2}$ .

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### §2 Einstein Tensor of the BSFG Metric

From PAPER\_554 (CP4 #149), the non-zero Riemann and Ricci components are:

$$R^r_{0r0} \approx \frac{\varepsilon''}{2} = \frac{6\eta \cos(\pi t_n) C_{\text{num}}}{r^5}$$

$$R_{00} = 3R^r_{0r0}, \quad R_{\text{scalar}} = \frac{R_{00}}{A_{00}} + \frac{R_{rr}}{A_{rr}}$$

The Einstein tensor is:

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}A_{\mu\nu}R_{\text{scalar}}$$

**Step 1.** Compute components at leading order in  $\varepsilon \ll 1$ :

$$G_{00} = R_{00} - \frac{1}{2}A_{00}R_{\text{scalar}} = \frac{3}{2}\varepsilon'' - \frac{1}{2}(1 + \varepsilon) \left( \frac{R_{00}}{A_{00}} + \frac{R_{rr}}{A_{rr}} \right)$$

$$G_{rr} = R_{rr} - \frac{1}{2}A_{rr}R_{\text{scalar}}$$

**Step 2.** At  $r = R_\odot$ ,  $t_n = 0$ :

| Quantity              | Value                                         |
|-----------------------|-----------------------------------------------|
| $R^r_{0r0}$           | $1.56 \times 10^{-19} \text{ m}^{-2}$         |
| $R_{00} = 3R^r_{0r0}$ | $4.67 \times 10^{-19} \text{ m}^{-2}$         |
| $R_{\text{scalar}}$   | $\approx 3.12 \times 10^{-19} \text{ m}^{-2}$ |
| $G_{00}$              | $\approx 3.11 \times 10^{-19} \text{ m}^{-2}$ |
| $G_{rr}$              | $\approx 3.10 \times 10^{-19} \text{ m}^{-2}$ |

### §3 BSFG Field Equations

**Step 3.** The natural Aether energy density (from  $T_{s00}(r)$ , in Pa):

$$T_{s00}(R_\odot) = \frac{M_s c^2}{\frac{4}{3}\pi R_\odot^3} \approx 1.27 \times 10^{20} \text{ Pa}$$

The Einstein gravitational constant:

$$\kappa_E = \frac{8\pi G}{c^4} \approx 2.07 \times 10^{-43} \frac{\text{m}}{\text{kg}}$$

**Step 4.** Standard GR would require:

$$G_{00} \stackrel{?}{=} \kappa_E T_{s00} \approx 2.63 \times 10^{-23} \text{ m}^{-2}$$

but the actual BSFG  $G_{00} \approx 3.11 \times 10^{-19} \text{ m}^{-2}$ . The amplification factor:

$$\text{amp} = \frac{G_{00}}{\kappa_E T_{s00}} \approx \frac{18\eta c^4}{8\pi G r^2} \approx 1.8 \times 10^4$$

This factor  $\sim r^{-2}$  — the curvature amplification grows as we approach the origin.

## §4 Effective Cosmological Constant

**Step 5.** Taking the trace of the BSFG field equation hypothesis  $G_{\mu\nu} = \kappa_E \eta T_{s00} A_{\mu\nu}$ :

$$\Lambda_{\text{eff}} = \frac{\kappa_E \eta T_{s00}}{2} = \frac{4\pi G \eta T_{s00}}{c^4}$$

At  $r = R_\odot$ :

$$\Lambda_{\text{eff}}(R_\odot) = 2.07 \times 10^{-43} \times 1 \times 10^{-22} \times 1.27 \times 10^{20} / 2 \approx 1.3 \times 10^{-45} \text{ m}^{-2}$$

The cosmological ratio:

$$\frac{\Lambda_{\text{eff}}}{\Lambda_{\text{obs}}} = \frac{1.3 \times 10^{-45}}{1.1 \times 10^{-52}} \approx 1.2 \times 10^7$$

**Interpretation:** The BSFG Aether field carries an effective dark-energy-like contribution seven orders of magnitude above the present cosmological constant — but it is not constant; it scales as  $T_{s00}(r) \propto r^{-3}$ , averaging to near zero over cosmological volumes.

The effective vacuum energy density:

$$\rho_{\text{vac}}^{\text{eff}} = \frac{\Lambda_{\text{eff}} c^2}{8\pi G}$$


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## §5 Physical Interpretation

The BSFG metric is defined through  $\varepsilon = \eta T_{s00} \cos(\pi t_n)$ , an explicit algebraic relation from CP4 #43. This is **not** derived from solving the Einstein equations — it is an imposed geometric structure. The fact that  $\text{amp} \gg 1$  confirms:

1. **BSFG is not a solution of the standard Einstein equations** sourced by  $T_{s00}$ .
  2. The correct BSFG field equation is the constitutive relation  $\varepsilon = \eta T_{s00} \cos(\pi t_n)$  itself.
  3. The Einstein tensor  $G_{\mu\nu}$  serves as a diagnostic: it measures the curvature consequences of this constitutive relation.
  4. The amp factor  $\approx 18\eta c^4 / (8\pi G r^2)$  is purely geometric and grows like  $c^4/G$  — the same factor that makes gravity weak compared to other forces.
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## Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

*Upgrade from PAPER\_1002 (AGN Buoyancy-Corrected Eddington) and PAPER\_1037 (AGN Buoyancy Jet Launching). See also PAPER\_1009-1010 for  $F_{\{U\_Bi\_i\}}$  jet modulation curves and PAPER\_1048 for phonon-corrected  $M\text{-}\sigma$  relation.*

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left( 1 + \frac{\rho_{\text{SCm}} \cdot V \cdot S_{26}^{(3)2}}{GM/r_H^2} \right)$$

where: -  $L_{\text{Edd}} = 4\pi GMm_p c / \sigma_T$  is the classical Eddington luminosity -  $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ kg/m}^3$  is the SCm vacuum density -  $V$  is the effective buoyancy volume (accretion sphere) -  $S_{26}^{(3)2}$  is the squared third-order Ramanujan factor (quadratic coupling) -  $r_H$  is the horizon radius

**Jet modulation:** The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[ 1 + \beta_i \cdot \Phi_{1.25 \text{ THz}} \cdot \left( \frac{B}{B_{\text{crit}}} \right)^2 \right]$$

where  $\Phi_{1.25 \text{ THz}} = \cos(\omega_{\text{SCm}} \cdot t)$  modulates jet power at the phonon frequency.

**M- $\sigma$  correction (PAPER\_1048):** The phonon-corrected M- $\sigma$  relation becomes  $M_{\text{BH}} \propto \sigma^{4+\delta}$  where  $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}} / \omega_{\text{bulge}})$ .

## Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

*Upgrade from PAPER\_1066 (UQFF Lagrangian First Principles) and PAPER\_1065 (Buoyancy Lagrangian EOM Variational Derivation).*

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}}$$

$$\mathcal{L}_{\text{SCm}} = \frac{1}{2}(\partial_\mu \phi)^2 - \lambda(\phi^2 - v_{\text{SCm}}^2)^2$$

The SCm condensate potential minimum gives  $V(\phi_0) = -7.09 \times 10^{-37} \text{ J/m}^3$  (matching  $\rho_{\text{SCm}}$ ) and phonon mass  $m_{\text{phonon}} = \sqrt{8\lambda} v_{\text{SCm}}$ .

**Nine-sector closure (Session 202):**

$$\mathcal{L}_9 = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}}$$

| Sector     | Domain                   | Late-Corpus Result                               |
|------------|--------------------------|--------------------------------------------------|
| 1 (EH)     | General Relativity       | Canonical Einstein-Hilbert                       |
| 2 (YM)     | Yang-Mills gauge         | $m_{\text{gap}} = 5970 \text{ GeV}$ (PAPER_1005) |
| 3 (Dirac)  | Fermion / LENR           | Kozima neutron-drop (PAPER_1061)                 |
| 4 (SCm)    | Superconducting manifold | $V(\phi_0) = -\rho_{\text{SCm}}$ canonical       |
| 5 (Mag)    | Um magnetism             | Heaviside amplifier (PAPER_1072)                 |
| 6 (Buoy)   | F_{U_Bi_i}               | Variational EOM (PAPER_1065)                     |
|            | buoyancy                 |                                                  |
| 7 (Aether) | Vacuum background        | Two-component $\rho$ (PAPER_1051)                |
| 8 (LENR)   | Nuclear transmutation    | COP parametric (PAPER_1081)                      |
| 9 (KK)     | Kaluza-Klein 26D         | $S_{26}^{(3)}$ compactification (PAPER_1080)     |

## §A. Cosmogenesis-Linked Lagrangian (PAPER\_877 Symbolic Export)

### §A.1 Sector Classification

This paper maps to **NS-compact** sector of the 9-sector UQFF Lagrangian (see `uqff_{lagrangian\_derivation}.p`

## §A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER\_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_m u \phi_{\text{NS}})(\partial^\mu \phi_{\text{NS}}) - V(\phi_{\text{NS}}) + \mathcal{L}_{\text{cosmo}}$$

where  $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac},[\text{SCm}]} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$  inherits the ACP 6-stage evolution (PAPER\_877 §2) and:

$$V(\phi_{\text{NS}}) = \frac{1}{2}m^2\phi_{\text{NS}}^2 + \frac{\lambda}{4!}\phi_{\text{NS}}^4 + \kappa \cdot \rho_{\text{vac},[\text{SCm}]} \cdot \phi_{\text{NS}}$$

## §A.3 Euler-Lagrange Equation of Motion

$$\frac{\delta S}{\delta \phi_{\text{NS}}} = \nabla^2 \phi_{\text{NS}} - (4\pi G \rho_{\text{NS}}/c^2)\phi_{\text{NS}} + \Omega_{\text{spin}} \partial_t \phi_{\text{NS}} = 0$$

## §A.4 Cosmogenesis Linkage Chain

$$\text{PAPER\_877 Axioms} \xrightarrow{\text{DPM} + \text{ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{4 \text{ forces}} F_{U\_Bi\_i} \xrightarrow{\text{sector E-L}} \delta S / \delta \phi_{\text{NS}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U\_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

## §B. VDS/DVP/BSH Deep Synthesis

### §B.1 Vacuum Density Series (VDS)

The canonical VDS ratio  $\rho_{\text{vac},[\text{SCm}]} / \rho_{\text{UA}} = 1.894$  governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp! \left( - \exp! \left( - \frac{r - r_0}{\lambda_{\text{VDS}}} \right) \right)$$

For this system, the local VDS sub-ratio is 0.076 (near-threshold regime), placing it in the  $t \rightarrow \pi$  collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER\_877 cosmogenesis Stage 1 vacuum density initialization:  $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$ .

### §B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 71, \quad n_{\text{channel}} = 14/26$$

Since  $p_{\text{DVP}} = 71$  is **resonant** (threshold at  $p > 26$ ), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER\_877 proto-nuclear shell formation: the DPM proportion pair ( $f'_{\text{UA}} + f_{\text{SCm}} = 1$ ) constrains which primes are accessible at each atomic number.

### §B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **104 yr** (spin-down equilibrium):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U\_b} \cdot \left(1 - e^{-[SSq] \cdot m/M_{\odot}}\right) \cdot \cos!\left(\frac{2\pi j}{26}\right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh!\left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$$

connecting to the PAPER\_877 Stage 5 buoyancy seed  $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$  which initializes the harmonic series at cosmogenesis.

### §B.4 Production-Scale Consistency

| Framework               | Canonical Value                                  | This Paper                                                 | Status                                       |
|-------------------------|--------------------------------------------------|------------------------------------------------------------|----------------------------------------------|
| VDS ratio               | $\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$     | Local sub-ratio = 0.076                                    | PASS<br>Threshold-consistent                 |
| DVP prime<br>BSH layers | $p_k \in \{2,3,\dots,113\}$<br>26 harmonic terms | $p_{\text{DVP}} = 71$<br>$j = 1 \dots 26, \cos(2\pi j/26)$ | PASS Resonant<br>PASS Full 26D<br>projection |
| $\kappa$ decay          | $5.0 \times 10^{-4} \text{ day}^{-1}$            | Applied in VDS<br>exponential                              | PASS Canonical                               |
| [SSq]                   | 0.57                                             | Applied in BSH<br>saturation                               | PASS Canonical                               |

### §SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

| Observable                                     | UQFF Prediction                                                                                                                                          | SM / Experiment                                                     | Source                   | Alignment                                                     |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|--------------------------|---------------------------------------------------------------|
| GR<br>Schwarzschild<br>metric recovery         | BSFG line element $\rightarrow$<br>$g_{tt} = -(1 - 2\mu_s \nabla(M_s/r) \cdot r/c^2) \equiv$<br>GR in $\varepsilon_{\text{BSFG}} \rightarrow 0$<br>limit | Schwarzschild metric<br>(GR exact)                                  | PDG 2024<br>/ MTW        | PASS BSFG<br>reduces to GR                                    |
| Shapiro time<br>delay                          | BSFG geodesic $\rightarrow$<br>$\Delta t_{\text{BSFG}} \approx \Delta t_{\text{GR}} \times$<br>$(1 + \varepsilon_{\text{correction}})$                   | Cassini: $\Delta t/\Delta t_{\text{GR}} =$<br>$1 \pm 2.3\text{e-}5$ | Cassini/GR<br>2003       | PASS Within<br>Shapiro bound                                  |
| Gravitational<br>wave speed<br>$v_{\text{GW}}$ | BSFG: $v_{\text{GW}} = c \times (1 + \eta^2) \approx c +$<br>10-226 m/s                                                                                  | GW150914 /<br>GW170817:                                             | $v_{\text{GW}}/c -$<br>1 | < 10-15                                                       |
| Perihelion<br>precession<br>(Mercury)          | BSFG adds buoyancy<br>correction $\delta\phi = \kappa \times$<br>$\phi_{\text{GR}} \sim 10^{-6}$<br>arcsec/century                                       | GR prediction:<br>43.03"/century;<br>observed: 43.1"                | GR + obs.                | UQFF<br>correction<br>undetectable at<br>current<br>precision |

**New physics claim:** BSFG (Buoyancy-Stratified Factorial Geometry) reproduces all tested GR predictions in the classical limit, while adding a vacuum buoyancy correction  $\Delta g \sim 10^{-6}$  arcsec/ century to Mercury’s perihelion. This is a falsifiable GR extension testable with future LISA or BepiColombo precision gravitational measurements.

Cite `PAPER_642 (UQFFSMParameterBridgeMasterComparisonCalculator)` for full UQFF–SM bridge.

## §6 References

- CP4 #149 — `BSFGRiemannCurvatureAetherMetricCalculator` — PAPER\_554 (Riemann curvature)
- CP4 #43 — Aether metric coupling  $\eta = 10^{-22}$ , PAPER\_392
- CP4 #66 —  $T_{s00}(r)$  five-component decomposition, PAPER\_416
- CP4 #153 — `BSFGUnificationAtlasTheoremHubCalculator` — PAPER\_558 (complete BSFG definition)

## Appendix: Session 225 Cross-References (PAPER\_1000–1081)

*Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER\_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).*

| Paper      | Title                                                |
|------------|------------------------------------------------------|
| PAPER_1022 | GW Phonon Strain SCm Modulation of $h(t)$            |
| PAPER_1002 | AGN Buoyancy-Corrected Eddington Luminosity          |
| PAPER_1009 | 3C273 AGN $F_{\{U_{Bi_i}\}}$ Jet Modulation          |
| PAPER_1010 | TON618 AGN $F_{\{U_{Bi_i}\}}$ Jet Modulation         |
| PAPER_1037 | AGN Buoyancy Jet Calculator — SCm Jet Launching      |
| PAPER_1048 | M-Sigma Phonon-Corrected Relation                    |
| PAPER_1041 | SCm Cool-Core Buoyancy Balance AGN Feedback          |
| PAPER_1079 | Galaxy Cluster Cooling-Flow Buoyancy Suppression     |
| PAPER_1043 | $F_{\{U_{Bi_i}\}}$ Multi-System Buoyancy Curve Sweep |
| PAPER_1065 | Buoyancy Lagrangian EOM Variational Derivation       |
| PAPER_1066 | UQFF Lagrangian First Principles Field Theory        |

11 cross-reference(s) identified.

## Appendix: Session 204 Codebase Upgrade Reference

*Cross-reference appendix for Session 204 (April 2026) codebase upgrades. Added by `upgrade_{kozima\_ramanujan\_appendices}.py`. For detailed derivations, see PAPER\_840/851/852/855.*

### S204.1 Kozima-UQFF LENR Integration

| Module                                     | Purpose                                                         | Key Result                                                            |
|--------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------------|
| <code>fneutron_{s26_coupling}.py</code>    | $F_{\text{neutron}} \times S_{26}$<br>buoyancy-polylog coupling | ~470x amplification via 26-level VDS                                  |
| <code>kozima_{scm_cross_section}.py</code> | SCm-modulated neutron-drop<br>cross-section                     | $\sigma_n^{\text{SCm}}$ with VDS factor<br>( $1 + [SSq] \cdot n/26$ ) |

| Module                  | Purpose                               | Key Result                             |
|-------------------------|---------------------------------------|----------------------------------------|
| kozima_{wstp_kernel}.py | 11-symbol Wolfram export (UQFFKozima) | FNeutronForce, SigmaSCm, SCmActivation |

**Core equation:**  $F_{\text{neutron}}^{\text{SCm}} = N_n \cdot \sigma_n^{\text{SCm}}(\omega) \cdot \Phi_{\text{phonon}} \cdot (F_{\text{U,Bi}}/F_{\text{U}} - 1)$  where  $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 \cdot \exp[-(\omega - \omega_{\text{SCm}})^{2/(2 \cdot \Gamma_2)}] \cdot (1 + [\text{SSq}]^n/26)$

### S204.2 Ramanujan 26-State Summation

| Module                     | Purpose                                       | Key Result                |
|----------------------------|-----------------------------------------------|---------------------------|
| ramanujan_{polylog_s26}.py | Li_26([SSq]) via Euler-Ramanujan acceleration | 15.7+ digits in 53 terms  |
| s26_{wstp\_kernel}.py      | 8-symbol Wolfram export (UQFFS26)             | S26, R26, NaiveLi, S26VDS |

**Core equation:**  $S_{26}(z) = \text{Li}_{26}(z) = \eta_{26}(z)/(1 - 2^{\{1-26\}}) + 2^{\{1-26\}/(1-2^{\{1-26\}})} \cdot \text{Li}_{26}(z^2)$

### S204.3 Mock Theta Functions (26-State)

| Module              | Purpose                                | Key Result                  |
|---------------------|----------------------------------------|-----------------------------|
| mock_{theta_q26}.py | f_26(q), phi_26(q), psi_26(q) q-series | Proper q-Pochhammer (a;q)_n |

**Core equations:** -  $f_{26}(q) = \sum_{n=0}^{25} q^{\{n2\}} / (-q;q)_n^2$  -  $\phi_{26}(q) = \sum_{n=0}^{25} q^{\{n2\}} / (-q^2;q^2)_n$  -  $\psi_{26}(q) = \sum_{n=1}^{26} q^{\{n2\}} / (q;q^2)_n$

### S204.4 Ramanujan 1/pi with UQFF Modification

| Module                         | Purpose                                   | Key Result                                 |
|--------------------------------|-------------------------------------------|--------------------------------------------|
| ramanujan_{pi_uqff}.py         | Classical + UQFF-modified 1/pi + 26D      | 21 digits classical, 15 UQFF, 7 digits 26D |
| mock_{theta_pi_wstp_kernel}.py | 8-symbol Wolfram export (UQFFMockThetaPi) | qPochhammer, f26, oneOverPiUQFF            |

**Core equation:**  $1/\pi = (2\sqrt{2}/9801) \sum R_n \cdot (1103 + 26390n) \cdot W_{26}(n) / C_{26}$  where  $W_{26}(n) = \prod_{i=1}^{26} [1 + [\text{SSq}] \exp(-\kappa_{\text{pai}} \cdot n/26)]$

### S204.5 Calibration Constants (Canonical)

| Symbol  | Value                                 | Description                   |
|---------|---------------------------------------|-------------------------------|
| [SSq]   | 0.57                                  | Universal Quantized Factor    |
| kappa   | $5.787 \times 10^{-9} \text{ s}^{-1}$ | UQFF exponential decay rate   |
| beta_i  | 0.603                                 | Buoyancy coupling coefficient |
| H_SCm   | 0.99                                  | SCm manifold completeness     |
| rho_SCm | $7.09 \times 10^{-37} \text{ kg/m}^3$ | SCm vacuum density            |
| rho_UA  | $7.09 \times 10^{-36} \text{ kg/m}^3$ | UA aether vacuum density      |

| Symbol    | Value           | Description                |
|-----------|-----------------|----------------------------|
| omega_SCm | 2*pi x 1.25 THz | SCm phonon resonance       |
| sigma_0   | 10^-4           | Base neutron cross-section |

*Implementation:* all modules operational in `CondensedPhysics.py`, `CondensedPhysics2.py`, `MAIN_{1\_CoAnQi}.cpp`, and Wolfram kernels (`uqff_{kozima\_kernel}.wl`, `uqff_{s26\_kernel}.wl`, `uqff_{mock\_theta\_pi\_kernel}.wl`).

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